

[RO] Input Json Spec

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Unit Information in JSON Spec

Type	Unit	Information
Length	mm	Millimeter
Time	sec	Second
Weight	kg	Kilogram
Volume	cbm	Cubic Meter
Speed	km/h	Kilometer per hour

Data Type Information in JSON Spec

Type	Format	Example
string	"Letter"	"This is a string :)"
Time	HI(0-23):MM:SS	23:59:59
DateTime	YYYY-MM-DD HI(0-23):MM:SS	2023-03-02 22:10:00
double	double number	10.4454
Integer	Integer number	-4, 1, 2, 3
Array	Array Data	{"1", "Hello"}
Object	{ "key" : "value" }	{ "id" : "Example ID" }

ObjectArray	[{"key": "value1"}, {"key": "value2"}]	[{"id": "Example ID_1"}, {"id": "Example ID_2"}]
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Json Spec

1. Plan

Plan Info includes continent, dataRange, plan options and depot information.

Property	Mandatory / Optional	Type	Format	Default	Description	Format / Range / Unit
planId	Mandatory	string			Unique ID for simulation	
continent	Mandatory	string			Continental information needed to calculate the distance between each place. CRO is composed of information for calculating distance based on actual roads by continent. Therefore, the user must enter that information.	Continent information includes EU(Europe), AS(Asia), NA (North America), SA (South America), OC (Australia, Oceania)
dateRange	Mandatory	object			Date range for simulation. Solver will create a plan in date range.	
● from	Mandatory	string	date-time		Start Date Time for simulation.	As defined by full-date - RFC3339
● to	Mandatory	string	date-time		End Date Time for simulation.	As defined by full-date - RFC3339

options	Optional	object			Plan options for more detailed plan. These are the simulation run options. Set to carry out planning more precisely. For the contents, please refer to 2. Plan Option.	
depot	Mandatory	object			Depot information. - Depot is a place where stores all items and all truck starts pick items from here and deliver to customers. - Each Plan has only one depot policy. If user want multiple depot in simulation, please contact administrator.	
● latitude	Mandatory	number	double		Latitude of depot.	
● longitude	Mandatory	number	double		Longitude of depot.	
● locId	Optional	string			(Optional) Unique ID for depot. If not entered, the engine will generate an ID for itself.	
● zoneId	Optional	string			(Optional) Zone ID for depot	
● open Time	Optional	string	partial-time	00:00:00	(Optional) Opening time of depot. - All vehicles must depart and arrive between opening time and closing time of depot. - All vehicles can't arrive and depart before opening time of depot. If then, they must wait until next opening time.	HI:MM:SS
● close Time	Optional	string	partial-time	23:59:59	(Optional) Closing time of depot. - All vehicles must depart and arrive between opening time and closing time of depot. - All vehicles can't arrive and depart after closing time of depot. If then, they must wait until next opening time.	HI:MM:SS

● taskTime	Optional	integer	int32	0	(Optional) Task time for depot. ● Task time is relevant to item installation. For example, it would take time for picking items in depot and move them to trucks.	Second
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2. Plan Option

Default Options

Option	Description	Type	format	Default value	Value Range
trips	trip end option ● oneway: vehicles does not come back to depot ● roundtrip: vehicles must come back to depot.	string		"oneway"	"oneway", "roundtrip"
multirotation	multi rotation ● vehicles can return to depot serveral times users set and pick up more orders. ● for example, if users set multirotation 2, trip could be like that. ○ depot(start) → 1st Location→ 2nd Location → depot(2nd visit) → 3rd Location ● value must be an integer and greater than -1. ○ -1 : vehicles can return to depot multiple times (unlimited) ○ 0 : vehicles can't return to depot ○ 1>= : vehicles can return to depot designated multi rotation times.	integer	int32	0	-1,0,1,2...
distanceCalculate	the way to calculate distance for each location ● RO user OSRM for calculate distance for each location. ● If you set other value like "GreatCircle", RO will calculate the distance as a straight line.	string		"OSRM"	"OSRM", "GreatCircle"
defaultSpeed	default speed for all vehicles ● if default speed is set, this value is set to for all vehicles.				

waitInDepot	drivers can wait in a depot and depart in late according to an opening time of 1st location. ● for example, if opening time of depot and 1st location are 9am and 1pm respectively, ○ vehicle does not need to depart at 9 am, because vehicle will wait until opening time of location. ○ in case, it is better for vehicle to depart in late and arrive at 1pm in 1st location.	string		"N"	
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3. Order

Order Info summarize demand side and includes due date, location, running hour of customer, item dimension and so on.

Property	Mandatory / Optional	Type	Format	Default	Description	Format / Range / Unit
orderId	Mandatory	string			A unique id for order	
reqDate	Optional	string	date-time	Plan End Date Time	Request date for order	for example, 2017-07-21T17:32:28Z
latitude	Mandatory	number	double		Latitude of order.	
longitude	Mandatory	number	double		Longitude of order.	
duration	Optional	Integer	int32		(Optional) Duration for location. ● For example, users can apply time-consuming jobs like parking, moving time for items and so on.	Second
openTime	Optional	string	partial-time	"00:00:00"	(Optional) Opening time of customer. ● All vehicles must depart and arrive between opening time and closing time of customer. ● All vehicles can't arrive and depart before opening time of customer. If then, they must wait until next opening time.	HH:MM:SS

closeTime	Optional	string	partial-time	"23:59:59"	(Optional) Closing time of customer. - All vehicles must depart and arrive between opening time and closing time of customer. - All vehicles can't arrive and depart after closing time of customer. If then, they must wait until next opening time.	HH:MM:SS
vehicleFeature	Optional	array		"ALL"	(Optional) Limitation to vehicle feature For example, only specific vehicles can load specific items or enter specific customer's location.	Allow types are: (multiple designations possible) "ALL", "5T", "15T", "20T", "25T"
items	Mandatory	array			Item information	
itemId	Mandatory	string			Unique Id for item.	
weight	Mandatory	number	double		Item weight. weight unit code must be KG (kilogram)	kg
volume	Mandatory	number	double		Item Volume volume unit code must be CBM(cubic meter)	cbm
qty	Mandatory	integer	int32	1	Item Qty	

taskTime	Optional	Integer	int32	0	(Optional) Task time for item. Users input item installation time repectively. Currently total task time is 30, not 300 (task time is calculated by item type not quntity, if you want to make tasktime based by quantity, please contact us, we will provide an option) <table><thead><tr><th>itemId</th><th>qty</th><th>taskTime</th></tr></thead><tbody><tr><td>1001</td><td>10</td><td>10</td></tr><tr><td>1002</td><td>10</td><td>10</td></tr><tr><td>1003</td><td>10</td><td>10</td></tr></tbody></table>	itemId	qty	taskTime	1001	10	10	1002	10	10	1003	10	10	Second
itemId	qty	taskTime																
1001	10	10																
1002	10	10																
1003	10	10																

4. Vehicle

Vehicle Info summarize supply side and includes vehicle feature, driver skill, running hour, max weight and volume, dimension and so on.

Property	Mandatory/Optional	Type	Format	Default	Description	Format / Range / Unit
vehicleId	Mandatory	string			Unique Id for vehicle	
vehicleFeature	Mandatory	string		"ALL"	Vehicle feature information Some orders have an limiation to vehicle feature. For example, some vhicle feature is allowed to load specific items or enter specific locations.	Allow types are: (multiple designations possible) "ALL" , "5T", "15T", "20T", "25T"
workStartTime	Optional	string	partial-time	"00:00:00"	(Optional) Work start time of vehicle. All vehicles can move between workStartTime and workEndTime.	HI:MM:SS
workEndTime	Optional	string	partial-time	"23:59:59"	(Optional) Work end time of vehicle. All vehicles can move between workStartTime and workEndTime.	HI:MM:SS
maxWeight	Mandatory	number	double		Max load weight of vehicle.	kg
maxVolume	Mandatory	number	double		Max load volumn of vehicle.	cbm

speed	Optional	number	double		(Optional) Speed of vehicle. ● Speed unit code must be km/h. ● If the speed of vehicle is set 100, the vehicle can travel 100km in an hour.	km/h
maxStopCount	Optional	number	integer		(Optional) Max Stop Count of vehicle ● If the max stop count of vehicle is set 10, the vehicle can load orders by 10 times.	
maxDrivingDistance	Optional	number	integer		(Optional) Max Driving Distance of vehicle. ● Vehicle can't travel more than max driving distance. ● If max driving distance is set 300,000 meter, driver can travel by 300 kilometer.	meter
maxDrivingTime	Optional	number	integer		(Optional) Max Driving Time of vehicle. ● Vehicle can't travel more than max driving time. ● If max driving time is set 360,000 seconds, driver can travel by 10 hours.	seconds

Json Body Sample

```
{
  "planId": "SampleTest",
  "continent": "AS",
  "dateRange": {
    "from": "2023-03-24 00:00:00",
    "to": "2023-03-24 23:59:00"
  },
  "options": {
    "trips": "oneway",
    "multiRotation": "0",
    "driverRestTimeRatio": "0.15",
    "waitInDepot": "Y"
  },
  "depot": [{
    "locId": "WARE_HOUSE",
    "latitude": "37.516250",
    "longitude": "127.100390",
    "taskTime": "0",
    "openTime": "09:00:00",
    "closeTime": "18:00:00"
  }],
  "orders": [
    {
      "orderId": "DO_001",
      "locId": "LOC_001",
      "dueDate": "2023-03-24 18:00:00",
      "customerId": "COMPANY_ID",

```



```
"latitude": "37.541000",
"longitude": "126.932000",
"openTime": "09:00:00",
"closeTime": "18:00:00",
"vehicleFeature": [
  "ALL"
],

"items": [
  {
    "itemId": "ITEM_001",
    "weight": "1",
    "volume": "1",

    "qty": "10",
    "taskTime": "10"
  },
  {
    "itemId": "ITEM_002",
    "weight": "2",
    "volume": "2",

    "qty": "10"
  }
],
{
  "orderId": "DO_002",
  "dueDate": "2023-03-24 18:00:00",
  "latitude": "37.485000",
  "longitude": "126.895000",
  "taskTime": "300",
  "openTime": "09:00:00",
  "closeTime": "18:00:00",
  "vehicleFeature": [
    "15T",
    "5T"
  ],

  "items": [
    {
      "itemId": "ITEM_003",
      "weight": "1",
      "volume": "1",

      "qty": "20",
      "taskTime": "5"
    }
  ],
{
  "orderId": "DO_003",
  "dueDate": "2023-03-24 13:00:00",
  "latitude": "36.311000",
  "longitude": "126.919000",
  "taskTime": "300",
  "openTime": "09:00:00",
  "closeTime": "18:00:00",
  "vehicleFeature": [
    "ALL"
  ],

  "items": [
    {
      "itemId": "ITEM_004",
      "weight": "5",
      "volume": "30",

      "qty": "1"
    }
  ]
}
```

```
}
],
"vehicles": [
  {
    "vehicleId": "V_001",
    "vehicleFeature": "15T",
    "driverSkill": "INSTALL",
    "maxWeight": "10.0",
    "maxVolume": "30.0",

    "workStartTime": "00:00:00",
    "workEndTime": "23:59:59"
  },
  {
    "vehicleId": "V_002",
    "vehicleFeature": "15T",
    "driverSkill": "ALL",
    "maxWeight": "20.0",
    "maxVolume": "20.0",

    "workStartTime": "00:00:00",
    "workEndTime": "23:59:59"
  },
  {
    "vehicleId": "V_003",
    "vehicleFeature": "15T",
    "driverSkill": "ALL",
    "maxWeight": "50.0",
    "maxVolume": "50.0",

    "workStartTime": "00:00:00",
    "workEndTime": "23:59:59"
  }
]
}
```